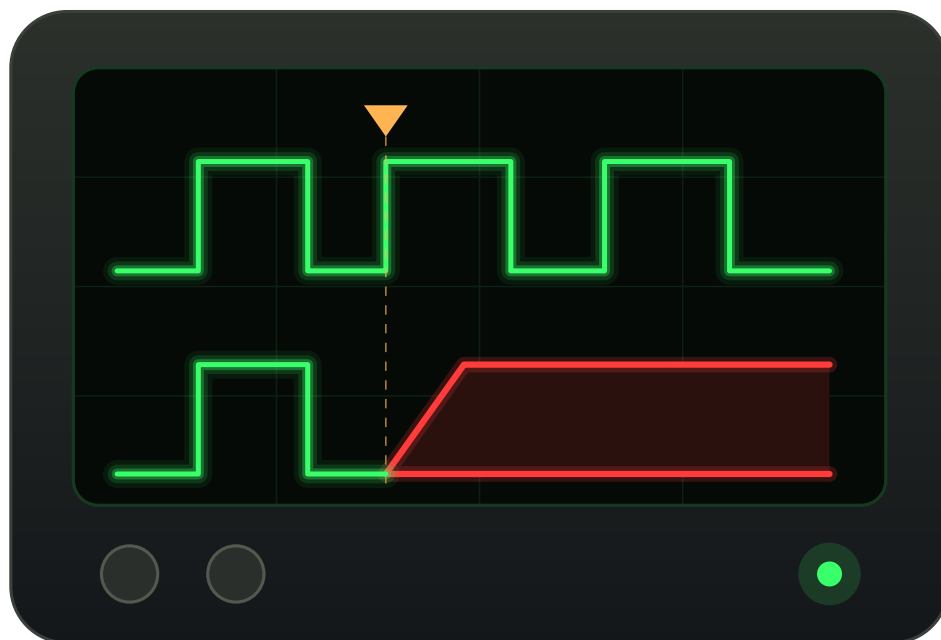




LibreILA

LibreILA User Manual

Yash Paritkar ([github/yashparitkar](https://github.com/yashparitkar))

July 2026

Document History

Version	Date	Comments
0.0	25 July 2026	Document created.

Contents

Document history	2
1 Introduction	5
2 Generating configuration	6
3 Creating a generic logic analyzer from LibreILA	7
4 Using the CLI interface of LibreILA	8
5 Using the GUI interface of LibreILA	9

List of Figures

List of Tables

List of Acronyms

AXI	Advanced eXtensible Interface
CDC	Clock Domain Crossing
FPGA	Field Programmable Gate Array
GUI	Graphical User Interface
HDL	Hardware Description Language
ILA	Integrated Logic Analyzer
IP	Intellectual Property
LSRAM	Large Static Random Access Memory
OHL	Open Hardware Licence
RTL	Register Transfer Level
UART	Universal Asynchronous Receiver Transmitter
VCD	Value Change Dump

1 Introduction

This document describes the design and implementation of the Integrated Logic Analyzer (ILA).

2 Generating configuration

Abstract

If same sampling clock speed is ok for multiple channels, a single ILA can be used to sample multiple channels otherwise multiple ILAs can be used to sample different channels at different speeds.

3 Creating a generic logic analyzer from LibreILA

4 Using the CLI interface of LibreILA

5 Using the GUI interface of LibreILA